

Gradebook with Class Statistics

Subject: C Programming

Technology Stack: C, VS Code

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Introduction

Records & Manages

A gradebook system efficiently records and manages student marks and grades.

Evaluates Performance

Essential for evaluating student academic performance and progress.

Automated Statistics

This project aims to automate grade calculation and class statistics generation.



Objectives



Input Student Marks

Input marks for multiple students, with a range of 0–100.



Assign Grades

Assign grades based on predefined criteria for each mark range.



Calculate Class Average

Determine the overall average marks for the entire class.



Identify Top Score

Find and display the highest individual score achieved in the class.



Display Grade Distribution

Present a summary of how grades are distributed among students.

[illegible]

User Input with Loops

Utilise loops for efficient input of marks for all students.

Conditional Grade Evaluation

Implement conditional statements (if/else) to accurately evaluate and assign grades.

Variables & Counters

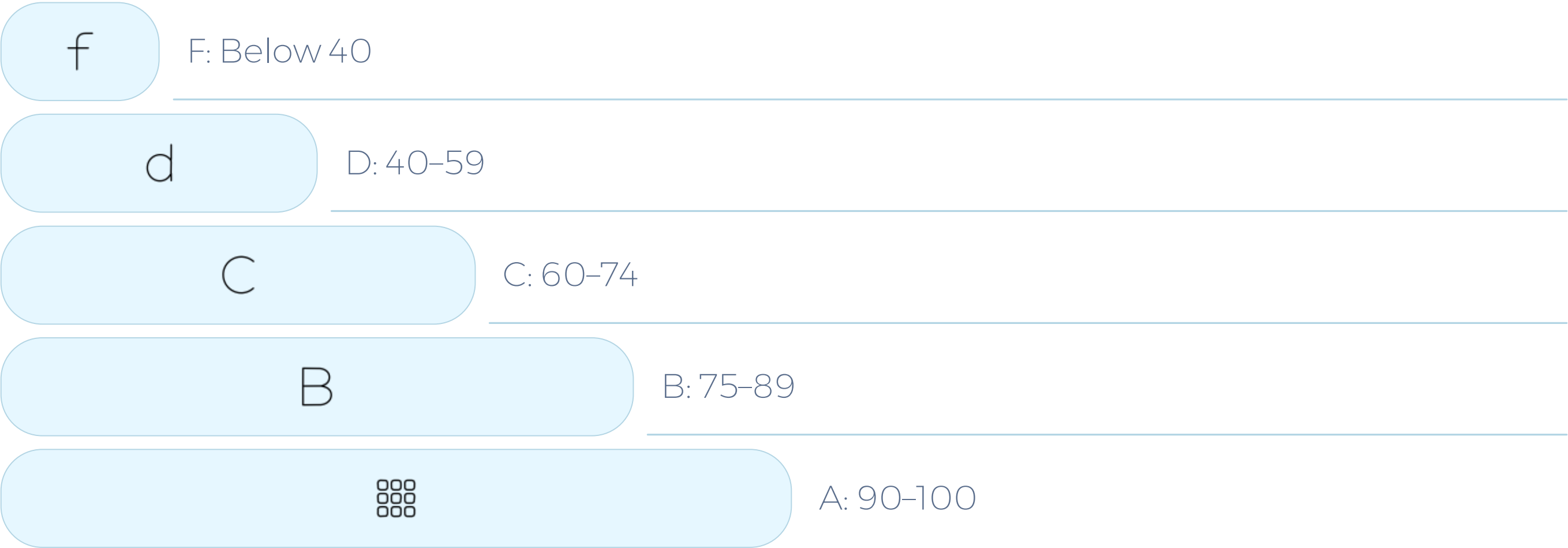
Employ variables and counters to store data and facilitate calculations.

Input Validation

Incorporate validation to ensure marks entered are strictly between 0 and 100.

Grade Criteria

Each subject is graded out of a total of 100 marks.



Output & Results



Average Marks

Clearly displays the calculated average marks for the entire class.

Top Score

Highlights the highest individual score achieved by any student.

Grade Distribution Chart

Presents a text-based summary of grade distribution (e.g., number of students with A, B, C grades).



Conclusion & Future Enhancements



Project Summary

Successfully automates grading and provides insightful class statistics.



Learning Outcomes

Enhanced understanding of loops, conditionals, and input validation in C programming.



Future Scope

- Menu-driven program for improved user experience.
- File handling to save and load student data persistently.
- Structure-based implementation for better data organisation.



GIT HUB REPOSITORY LINK

https://github.com/tushargiri05/C_MINIPROJECT



Thank You

For your attention and consideration.